

MID-WAY, TX, USA

WMO: 720299

Lat: 32.456N

Lon: 96.913W

Elev: 228

StdP: 98.61

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.2	(c) -2.3	(d) -11.4	(e) 1.5	(f) 0.7	(g) -9.1	(h) 1.8	(i) 2.7	(j) 11.9	(k) 12.2	(l) 11.1	(m) 11.4	(n) 3.2	(o) 350	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.1	(c) 37.9	(d) 22.6	(e) 37.1	(f) 22.6	(g) 35.8	(h) 22.8	(i) 25.5	(j) 31.9	(k) 25.1	(l) 31.4	(m) 24.7	(n) 30.8	(o) 3.5	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.6	26.6	23.7	19.0	26.6	23.1	18.3	26.5	79.2	31.7	77.5	31.4	75.9	30.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.9	8.0	-8.5	39.6	2.4	1.4	-10.2	40.7	-11.7	41.5	-13.0	42.3	-14.8	43.4		
			-8.8	25.9	2.6	1.1	-10.6	26.7	-12.1	27.3	-13.6	27.9	-15.5	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	8.1	9.9	14.6	18.2	22.6	27.2	28.9	29.2	25.8	19.5	13.6	8.8
	DBStd	8.66	5.49	5.79	5.13	4.11	3.32	2.15	2.08	2.53	3.13	4.73	5.18	5.12
	HDD10.0	301	101	68	19	2	0	0	0	0	0	2	24	84
	HDD18.3	1248	317	241	136	51	7	0	0	0	0	44	157	296
	CDD10.0	3553	43	66	161	249	391	515	584	596	473	296	131	48
	CDD18.3	1459	1	4	18	48	139	265	326	338	223	79	13	2
	CDH23.3	16055	11	30	115	348	1167	2911	4020	4296	2352	714	80	12
CDH26.7	7991	0	2	13	70	402	1443	2200	2446	1152	247	13	2	
Wind	WSAvg	3.4	3.8	4.0	4.2	4.0	3.5	3.2	2.9	2.6	2.5	3.0	3.4	3.4
Precipitation	PrecAvg	917	46	69	76	94	125	81	50	52	99	98	65	68
	PrecMax	1499	123	148	185	286	257	208	169	150	247	282	160	215
	PrecMin	499	0	4	7	4	28	5	2	2	21	8	10	9
	PrecStd	199	36	39	48	70	70	54	47	43	67	83	38	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	26.5	28.0	31.3	34.1	37.6	39.7	40.2	37.5	33.6	28.5	24.9
		MCWB	15.0	16.4	18.1	20.1	22.8	22.2	22.8	21.9	21.3	21.9	19.0	16.8
	2%	DB	22.1	23.9	26.3	28.3	32.2	36.0	37.7	38.5	35.9	31.5	25.7	22.1
		MCWB	14.6	16.9	17.7	19.7	21.8	22.3	22.6	22.8	22.0	21.6	18.0	16.5
	5%	DB	19.8	21.3	24.4	27.1	30.6	34.6	36.4	37.4	34.1	29.5	23.2	19.8
		MCWB	14.3	15.7	17.1	19.1	21.9	22.9	22.9	22.8	22.2	20.7	17.6	15.8
	10%	DB	17.4	19.6	22.5	25.3	29.0	33.0	35.1	36.1	32.7	27.5	21.6	17.5
		MCWB	13.1	15.1	16.6	18.6	21.4	23.0	23.3	22.9	21.9	19.1	16.9	13.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	20.4	20.7	23.0	25.1	26.2	26.2	25.8	25.3	24.3	22.2	20.3
		MCDB	20.9	23.0	24.6	26.6	29.7	32.4	33.0	32.6	31.1	28.9	25.0	21.8
	2%	WB	17.2	19.0	19.7	21.8	24.0	25.4	25.5	25.2	24.8	23.4	20.5	18.5
		MCDB	19.3	21.3	23.5	25.8	28.4	31.0	32.2	32.1	30.3	27.9	22.6	20.6
	5%	WB	15.7	17.4	18.9	20.9	23.3	24.8	25.0	24.8	24.2	22.6	19.2	16.5
		MCDB	18.6	20.2	22.5	24.3	27.8	30.3	31.6	31.7	29.3	26.9	22.0	18.4
	10%	WB	13.3	15.6	17.9	20.1	22.6	24.2	24.6	24.4	23.6	21.6	17.9	14.6
		MCDB	16.9	18.7	21.2	23.9	27.0	29.6	30.9	31.3	28.7	25.6	20.8	17.1
Mean Daily Temperature Range	5% DB	MDBR	11.1	10.9	11.4	11.6	10.7	11.2	11.7	12.1	12.1	12.1	11.3	10.7
		MCDBR	14.3	13.7	12.8	12.5	12.4	13.0	13.3	13.7	13.5	13.3	12.6	13.8
		MCWBR	8.7	8.3	6.8	6.2	4.4	3.3	3.0	3.1	3.8	5.3	7.0	8.9
	5% WB	MCDBR	11.9	11.7	10.5	10.0	10.0	10.2	11.2	11.6	11.0	11.2	10.1	11.1
		MCWBR	8.9	8.4	6.6	6.1	4.5	3.5	3.2	3.0	3.7	5.3	7.5	9.4
Clear-Sky Solar Irradiance	taub	0.313	0.329	0.354	0.394	0.422	0.433	0.437	0.437	0.398	0.357	0.337	0.318	
	taud	2.506	2.467	2.414	2.324	2.303	2.304	2.332	2.314	2.375	2.476	2.473	2.495	
	Ebn at Noon	911	928	925	897	869	855	849	846	868	885	872	883	
	Edh at Noon	86	99	112	128	132	131	127	128	115	97	88	83	
All-Sky Solar Radiation	RadAvg	2.89	3.36	4.46	5.41	5.83	6.56	6.70	6.16	5.18	4.08	3.06	2.55	
	RadStd	0.28	0.54	0.36	0.41	0.51	0.60	0.46	0.41	0.43	0.60	0.39	0.33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page